# Results pack Q2: Cumulative fluid balance and acute kidney injury

*FLUID-ICU is a SIMULATED dataset made for teaching. No real patients. Neudata | Foundations of Clinical Research, Level 1*

**SIMULATED DATA: NOT REAL PATIENTS.** FLUID-ICU was generated by computer for this course. Write your manuscript as a training exercise and state clearly that the data are simulated.

## 1. The question

**In adults with sepsis in the ICU, is a higher cumulative fluid balance by day 3 associated with acute kidney injury?**

## 2. Design and data

- Design: multicentre prospective cohort study (observational).
- Setting: 4 ICUs (ICU A-D), consecutive adults with sepsis, enrolment January 2023 to December 2024.
- Exposure: cumulative fluid balance at day 3 in ml/kg, analysed per 10 ml/kg (continuous); secondary: positive vs negative balance.
- Outcome: acute kidney injury (KDIGO stage >= 1, maximum stage during ICU days 1-7; aki_any). Secondary: AKI stage 2-3; renal replacement therapy (descriptive).
- Analysis sample: 618 patients with a day-3 balance; adjusted models 527 complete cases.
- Pre-specified confounders (SAP): age, SOFA score, lactate, vasopressor, mechanical ventilation, chronic kidney disease.
- Software: R 4.6 (packages gtsummary, survival, mice). Scripts: Course/Data/R/.

Participant flow: 620 patients enrolled -> 2 without a day-3 fluid balance (weight or output missing after cleaning) -> 618 in the exposure analysis -> 91 without lactate (or age) -> 527 in the complete-case adjusted model.

## 3. Table 1: who was studied

**Table 1. Baseline characteristics by fluid-balance group**

| Characteristic | Overall N = 618[1] | Negative balance N = 245[1] | Positive balance N = 373[1] |
|---|---|---|---|
| **Age, years** | 62.6 (15.2) | 64.6 (15.0) | 61.4 (15.2) |
| Missing | 1 | 1 | 0 |
| **Sex** | | | |
| Female | 243 (39%) | 102 (42%) | 141 (38%) |
| Male | 375 (61%) | 143 (58%) | 232 (62%) |
| **Weight, kg** | 64.7 (11.9) | 65.1 (12.4) | 64.4 (11.6) |
| **Diabetes** | 158 (26%) | 70 (29%) | 88 (24%) |
| **Chronic kidney disease** | 64 (10%) | 28 (11%) | 36 (9.7%) |
| **Heart failure** | 97 (16%) | 54 (22%) | 43 (12%) |
| **Hypertension** | 282 (46%) | 124 (51%) | 158 (42%) |
| **Source of infection** | | | |
| Respiratory | 267 (43%) | 101 (41%) | 166 (45%) |

| Characteristic | Overall N = 618[1] | Negative balance N = 245[1] | Positive balance N = 373[1] |
|---|---|---|---|
| Urinary | 126 (20%) | 54 (22%) | 72 (19%) |
| Abdominal | 106 (17%) | 46 (19%) | 60 (16%) |
| Skin/soft tissue | 57 (9.2%) | 18 (7.3%) | 39 (10%) |
| Other | 62 (10%) | 26 (11%) | 36 (9.7%) |
| **SOFA score** | 7 (5, 9) | 6 (4, 8) | 8 (6, 10) |
| **APACHE II score** | 19 (15, 24) | 17 (12, 21) | 20 (17, 25) |
| Missing | 6 | 3 | 3 |
| **Lactate, mmol/L** | 2.6 (1.7, 3.7) | 2.2 (1.4, 3.0) | 2.9 (1.9, 4.0) |
| Missing | 91 | 50 | 41 |
| **Mean arterial pressure, mmHg** | 73 (66, 80) | 76 (70, 84) | 71 (65, 78) |
| Missing | 4 | 1 | 3 |
| **Vasopressor on admission** | 266 (43%) | 61 (25%) | 205 (55%) |
| **Mechanical ventilation** | 313 (51%) | 97 (40%) | 216 (58%) |

[1]Mean (SD); n (%); Median (Q1, Q3)

*Mean (SD); n (%); median (Q1, Q3). Missing = number of patients without that value.*

## 4. Main results

- AKI (any stage) occurred in 302/620 (48.7%) of all patients; RRT in 79/620 (12.7%).
- AKI: positive balance 230/373 (61.7%) vs negative balance 71/245 (29.0%).
- AKI rose with each balance category: <= 0: 29%; >0 to 20: 54%; >20 to 40: 63%; > 40: 75%.
- Crude OR per 10 ml/kg more balance: 1.37 (95% CI 1.27 to 1.47).
- Adjusted OR per 10 ml/kg: **1.27 (95% CI 1.16 to 1.40)**, p <0.001.
- Positive vs negative balance: crude OR 3.94 (95% CI 2.80 to 5.60); adjusted OR 2.70 (95% CI 1.76 to 4.15).

**Table 2. Maximum KDIGO AKI stage by fluid-balance group**

| AKI stage | Negative balance | Positive balance |
|---|---|---|
| KDIGO stage 0 | 174 (71%) | 143 (38%) |
| KDIGO stage 1 | 29 (12%) | 68 (18%) |
| KDIGO stage 2 | 24 (10%) | 54 (14%) |
| KDIGO stage 3 | 18 (7%) | 108 (29%) |

**Table 3. AKI by category of cumulative fluid balance at day 3**

| Cumulative balance day 3, ml/kg | n | Any AKI, n (%) | AKI stage 2-3, % | RRT, % |
|---|---|---|---|---|
| <= 0 | 245 | 71 (29%) | 17.1 | 4.1 |
| >0 to 20 | 166 | 90 (54%) | 30.7 | 11.4 |

| Cumulative balance day 3, ml/kg | n | Any AKI, n (%) | AKI stage 2-3, % | RRT, % |
|---|---|---|---|---|
| >20 to 40 | 130 | 82 (63%) | 52.3 | 21.5 |
| > 40 | 77 | 58 (75%) | 55.8 | 28.6 |

**Table 4. Logistic regression for AKI (adjusted for age, SOFA, lactate, vasopressor, ventilation, CKD)**

| Model | n | OR (95% CI) | p |
|---|---|---|---|
| Balance per 10 ml/kg, crude | 618 | 1.37 (1.27 to 1.47) | <0.001 |
| Balance per 10 ml/kg, adjusted | 527 | 1.27 (1.16 to 1.40) | <0.001 |
| Positive vs negative balance, crude | 618 | 3.94 (2.80 to 5.60) | <0.001 |
| Positive vs negative balance, adjusted | 527 | 2.70 (1.76 to 4.15) | <0.001 |
| Balance per 10 ml/kg, adjusted (outcome AKI stage 2-3) | 527 | 1.23 (1.12 to 1.35) | <0.001 |

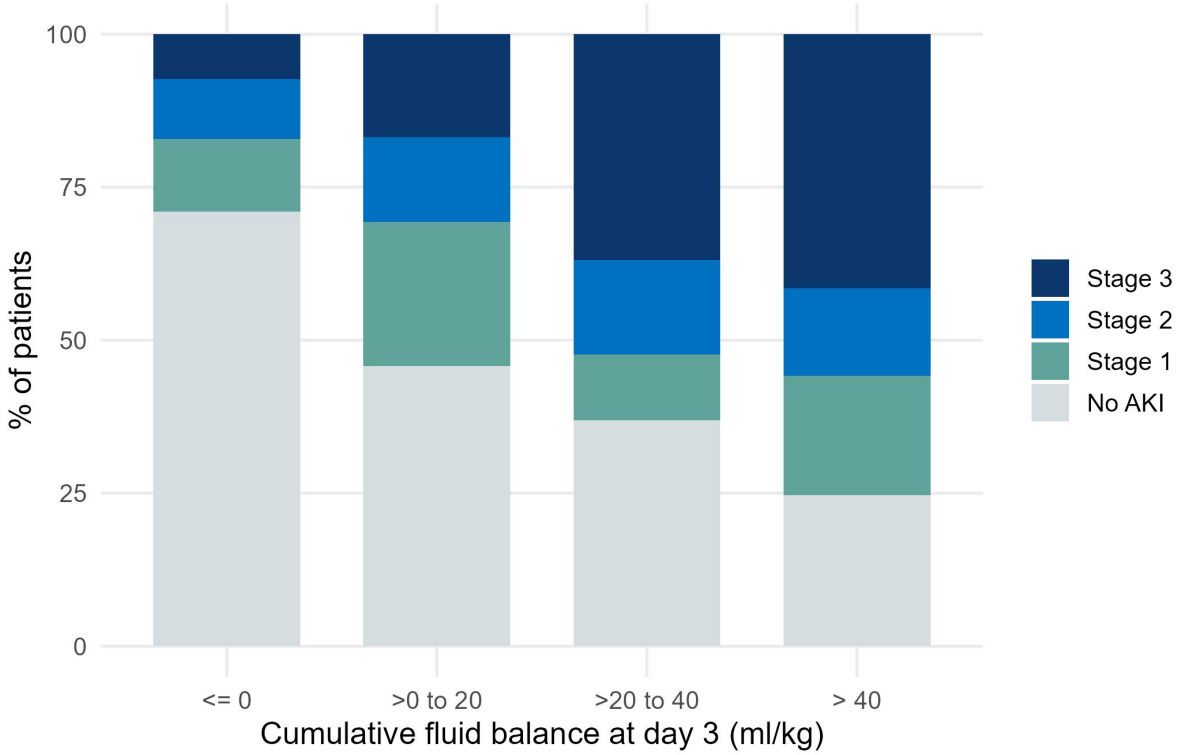



*Figure 1. Maximum AKI stage by category of cumulative fluid balance*

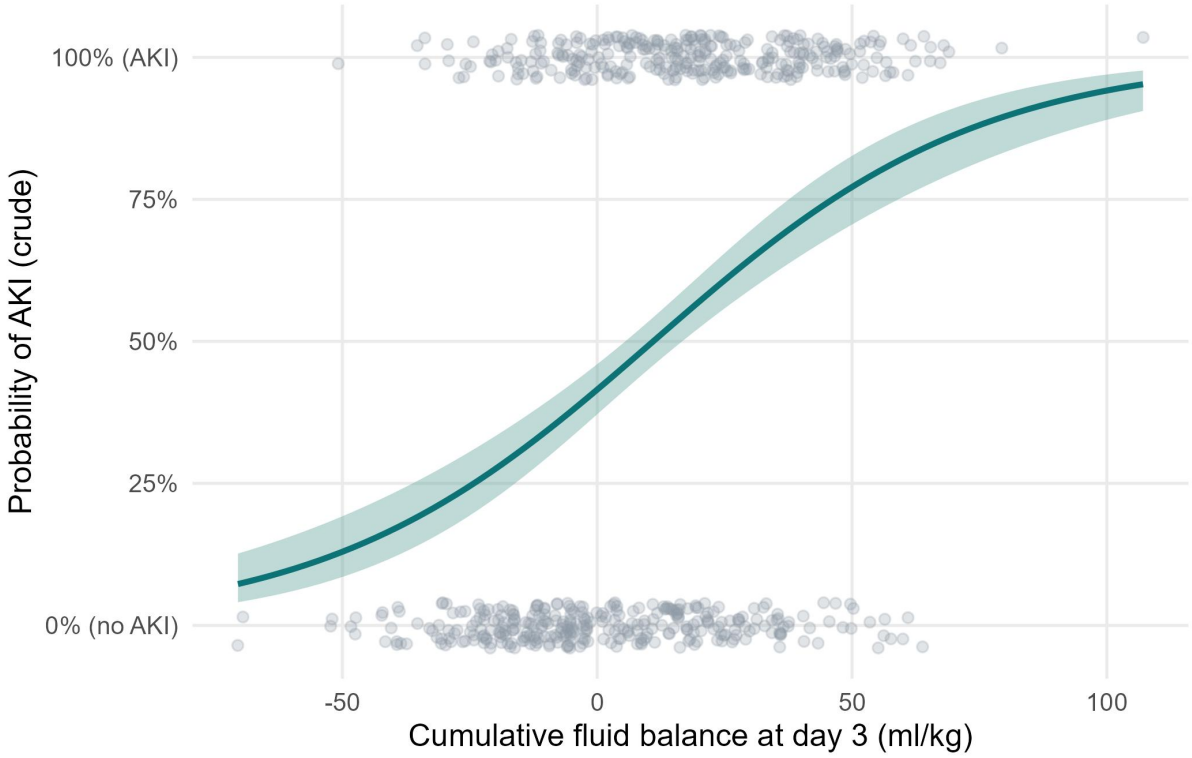



*Figure 2. Crude probability of AKI across the range of fluid balance (logistic curve, 95% band)*

## 5. What this means (plain language)

- AKI was more common the more positive the fluid balance: a clear 'dose-response' pattern.
- After adjustment, each extra 10 ml/kg (about 650 ml for a 65 kg adult) was associated with about 27% higher odds of AKI.
- Using the continuous balance keeps more information than positive/negative and shows the gradient.

## 6. What this does NOT mean

- It does NOT show that fluid damages the kidney. The arrow may point the other way: AKI reduces urine output, and low output makes the balance positive (reverse causation).
- We do not know whether AKI started before or after the fluid accumulated. The data record the maximum stage over days 1-7.
- An OR of 1.27 per 10 ml/kg is not a risk: with AKI this common (about 49%), odds ratios overstate risk ratios.

## 7. Limitations to discuss (hints)

- Timing: exposure (days 1-3) and outcome (days 1-7) overlap.
- Residual confounding by severity and nephrotoxic drugs (not recorded).
- KDIGO staging may be based on creatinine, which fluid dilutes, so AKI may be under-diagnosed in patients with large positive balances.
- Complete-case analysis (lactate missing in 15%).

## Which STROBE items this pack helps you write

- Item 6-7: participants and variable definitions (KDIGO)
- Item 12: methods incl. continuous exposure scaling (per 10 ml/kg)
- Item 14: descriptive data (Table 1)
- Item 15: outcome data (Tables 2-3)
- Item 16: crude and adjusted ORs with 95% CI; category boundaries reported (Table 3-4)
- Item 19: limitations (reverse causation and timing)

- Item 17: secondary outcome AKI stage 2-3